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MVTec AD — A Comprehensive Real-World Dataset for Unsupervised Anomaly Detection

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Abstract

The detection of anomalous structures in natural image data is of utmost importance for numerous tasks in the field of computer vision. The development of methods for unsupervised anomaly detection requires data on which to train and evaluate new approaches and ideas. We introduce the MVTec Anomaly Detection (MVTec AD) dataset containing 5354 high-resolution color images of different object and texture categories. It contains normal, i.e., defect-free, images intended for training and images with anomalies intended for testing. The anomalies manifest themselves in the form of over 70 different types of defects such as scratches, dents, contaminations, and various structural changes. In addition, we provide pixel-precise ground truth regions for all anomalies. We also conduct a thorough evaluation of current state-of-the-art unsupervised anomaly detection methods based on deep architectures such as convolutional autoencoders, generative adversarial networks, and feature descriptors using pre-trained convolutional neural networks, as well as classical computer vision methods. This initial benchmark indicates that there is considerable room for improvement. To the best of our knowledge, this is the first comprehensive, multi-object, multi-defect dataset for anomaly detection that provides pixel-accurate ground truth regions and focuses on real-world applications.

1. Introduction

Humans are very good at recognizing if an image is similar to what they have previously observed or if it is something novel or anomalous. So far, machine learning systems, however, seem to have difficulties with such tasks.

There are many relevant applications that must rely on unsupervised algorithms that can detect anomalous regions. In the manufacturing industry, for example, optical inspection tasks often lack defective samples or it is unclear what

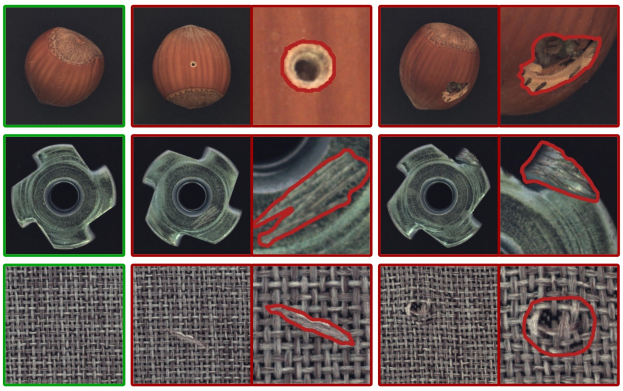


Figure 1: Two objects and one texture from the MVTec AD dataset. For each of them, one defect-free image and two images that contain anomalies are displayed. Anomalous regions are highlighted in close-up figures together with their pixel-precise ground truth labels. The dataset contains objects and textures from several domains and covers various anomalies that differ in attributes such as size, color, and structure.

kinds of defects may appear. In active learning systems, structures that are identified as anomalous might indicate the necessity of including a specific image for training. Therefore, it is not surprising that recently a significant amount of interest has been directed towards novelty detection in natural image data using modern machine learning architectures. A number of algorithms have been proposed that test whether a network is able to detect if new input data matches the distribution of the training data. Many of these algorithms, however, focus on classification settings in which the inlier and outlier distributions differ significantly. This is commonly known as outlier detection or one-class-classification. A common evaluation protocol is to arbitrarily label a number of classes from existing object